The Inheritance Law on the Cycle

Paper 3. Exact transport of the divisor census, and of weights, over a sieve cycle

---

Abstract

We study the odd sieve over one full period $\prod q$ of its lines, in the cells of Paper 2. Each cell is in one of four states, by which of its two members survive.

Main result (Theorem 1). After the lines $5 \le q \le z$ have acted, the four state shares are exactly

$$(a,b,c,d) = (P_2,\ P_1-P_2,\ P_1-P_2,\ 1-2P_1+P_2), \qquad P_1 = \prod_{5\le q\le z}\Big(1-\frac{1}{q}\Big),\quad P_2 = \prod_{5\le q\le z}\Big(1-\frac{2}{q}\Big).$$

The four counts are a census, not a simulator: two cells can end in the same state by different histories.

Refinements, all exact on the cycle. - Counting also how many lines divide each cell, the census has generating function $\prod_q \big((q-2)+xu+xv\big)$ (Theorem 2). So $\sum f(\Omega_{\le z})$ is exact for any f (Corollary 3). - Marking each line by the bin of its size transports the per-bin census exactly (Corollary 4); eight bins reproduce Richert's logarithmic weight to 0.4%. - Giving each integer to its smallest striking line splits the strip into disjoint layers, so the survivor count needs no inclusion–exclusion (Theorem 3). - The pair correlation obeys $C_{\text{new}}(h) = K_q(h)C_{\text{old}}(h)$ with $K_q(h) \in \{q-2,\ q-3,\ q-4\}$ (Theorem 4). This is the Hardy–Littlewood singular series, reached by a direct count.

Everything here is exact on the full cycle. The passage to a short window is Paper 4, and that is where the losses are.

Keywords: sieve of dimension two, Mertens products, weighted sieves, Richert weights, generating functions.

MSC 2020: 11N35, 11N05, 11Y16.

How to read the claims in this paper. Statements set as Theorems, Propositions and Corollaries are proved, and the proofs are given. Everything else falls into two kinds, and we try to keep them apart. A *measurement* is a computation over a stated finite range; it is labelled with that range, and it supports a claim about that range only. A *reading* is our own judgement about what a measurement or a proof appears to mean, and we mark it as ours rather than stating it as established.

---

1. Setting

We use the coordinates of Paper 2 without change. $L_p(k) = p(p+2k)$ is the line of p, beginning at p^2; L_3 is the coordinate grid, and the odd integers compress to cells $C_b = (6b-1,\ 6b+1)$. Two facts from Paper 2 are used throughout and nothing else is:

- **[P2, Thm 2] (the cell map).** With $p = 6a+\sigma$, one has $p(6b+\varepsilon) = 6(pb+a\varepsilon)+\sigma\varepsilon$, so the line p sends the two members of C_b into the cells $pb-a$ and $pb+a$. In cell coordinates each branch steps by p, not by $6p$.

- **[P2, Thm 3].** The two positions a line closes are $j \equiv \pm 6^{-1} \pmod{p}$, and they are always distinct; so **exactly two** of every p cells lose a member, one on each rail.

The second is the only input the whole of §2–§4 needs. Everything below is a consequence of "of the p copies of any cell, exactly two are struck, one per rail."

Throughout, z is the sieve depth, the cycle modulus in the cell index is $\prod_{5\le q\le z} q$ (**not** $6\prod q$ — an off-by-a-factor that is easy to make), and $\Omega_{\le z}$ denotes the number of lines up to z dividing either member of a cell, i.e. the inheritance depth. On this cycle a line is seen at most once per member, so this counts distinct lines; §4.1 says how to recover the multiplicity.

---

2. The four-state law and its closed solution

2.1 The update

Counting cells of each type over a full cycle, the entry of line p gives

$$A' = (p-2)A, \quad B' = A + (p-1)B, \quad C' = A + (p-1)C, \quad D' = B + C + pD. \tag{2.1}$$

Reason. Of the p copies of an NN cell, one has its lower member struck and one its upper, so $p-2$ remain and two migrate. In NO one member is already dead, so only the live member can be hit and one copy migrates to OO. OO is invariant.

Verification (exact rational arithmetic; the cycle modulus in the cell index is $\prod q$, **not** $6\prod q$):

lines	M	(A, B, C, D)	predicted
$\{5\}$	5	$(3, 1, 1, 0)$	—
$\{5, 7\}$	35	$(15, 9, 9, 2)$	✓
$\{5, 7, 11\}$	385	$(135, 105, 105, 40)$	✓
$\{5, 7, 11, 13\}$	5,005	$(1485, 1395, 1395, 730)$	✓

The full cycle state — which grows like $\prod q$ — is thus carried by four integers.

2.2 Theorem 1 (conserved quantity and closed solution)

Write $a = A/M$ and so on.

> **Theorem 1.** The quantity $a + b$ is form-invariant, $(a+b)' = (1 - 1/p)(a+b)$, and consequently $\boxed{(a, b, c, d) = (P_2, P_1 - P_2, P_1 - P_2, 1 - 2P_1 + P_2)}$ where $P_1 = \prod_{5\le q\le z}(1 - 1/q)$ and $P_2 = \prod_{5\le q\le z}(1 - 2/q)$, the products running to the sieve depth z of §1.

Proof. From (2.1) in normalised form,

$$a' + b' = \left(1 - \tfrac{2}{p}\right)a + \left(1 - \tfrac{1}{p}\right)b + \tfrac{a}{p} = a\left(1 - \tfrac{2}{p} + \tfrac{1}{p}\right) + \left(1 - \tfrac{1}{p}\right)b = \left(1 - \tfrac{1}{p}\right)(a + b).$$

Since $a+b = 1$ initially, $a+b = P_1$. Also $a' = (1-2/p)a$ with $a = 1$ initially gives $a = P_2$. Symmetry gives $b = c$, and $a + b + c + d = 1$ supplies d. ■

Verification. All three identities hold with zero error in exact rational arithmetic for every p from 5 to 199.

p	a	b	$a+b$	P_1	P_2
7	0.428571	0.257143	0.685714	0.685714	0.428571
31	0.186275	0.272282	0.458556	0.458556	0.186275
199	0.085574	0.226109	0.311684	0.311684	0.085574

2.3 Asymptotics

Removing the factors at 2 and 3 (whose product is 1/3) from Mertens' theorem [1],

$$P_1 \sim \frac{C}{\log P}, \quad C = 3e^{-\gamma} = 1.6843785; \qquad P_2 \sim \frac{K}{\log^2 P}, \quad K = 12C_2e^{-2\gamma} = 2.4972872.$$

Verification. Iterating (2.1) to $P = 2 \times 10^8$: $a \log^2 P = 2.497274$; and $(1-d)\log P = 3.23809$, approaching the predicted $2C = 3.3687569$ from below, since $(1-d)\log P = 2P_1 \log P - P_2 \log P = 2C - K/\log P + o(1/\log P)$.

2.4 Structural reading

$$\frac{b}{a} \sim \frac{C}{K} \log P = 0.6744833 \log P.$$

Fully alive cells — the twin candidates — die like $1/\log^2 P$, whereas half-dead cells die only like $1/\log P$. The flow is always $NN \to NO/ON \to OO$, and the surviving population becomes overwhelmingly OO.

Methodological remark. Although (2.1) arises from the cell model alone, it dissolves completely into the two classical Mertens products and contains no new quantity. We record it as a fact about the law, not as a disappointment: it is the reason the law is *exact* rather than approximate.

2.5 The law is a census, not a simulator

Theorem 1 transports the four counts exactly, and it is easy to read more into that than it says. It does not say that the four states carry enough information to run the process, and they do not.

The counterexample is two cells. Take

$$C_4 = (23,\ 25), \qquad C_8 = (47,\ 49).$$

Both end in the same state — one member survives, the other is composite — so both are counted in the same class by Theorem 1 at every depth beyond 7. But their histories differ: $25 = 5^2$ is struck the moment L_5 enters, whereas $49 = 7^2$ is still **alive** at that stage and dies only when L_7 enters. So two cells with identical final states occupy different states at an intermediate depth, and the final label does not determine which.

Consequence, stated as a limitation. No automaton on the four states reproduces the history of individual cells. To recover that history one needs at least the *owner* (or death stage) of each dead member, which is strictly more information than the final state. **Inheritance depth is different information:** it records how many old lines divide either member, not which line killed a member first and not when that happened. Theorem 2 restores the depth distribution, which is enough for

the weighted censuses of §§3–4, but it does **not** turn the four-state census into a history simulator. Ownership is introduced separately in §5.

And it locates a second reading of the same fact. The four-state law is exact *because* it forgets: a linear update on four numbers can only exist if the update is independent of everything the four numbers do not record. Owner labels and spatial history are precisely what have to be forgotten for the four-number law to close. So exactness and blindness are two sides of the same compression here — which is the pattern this framework repeats, and the reason Paper 4 has something to measure at all.

3. Theorem 2: the refined law, with generating function

Theorem 1 records how many cells occupy each state. One may ask for more: the distribution of **inheritance depth** across states, where the depth j of a cell is the number of old lines dividing either of its members. This refinement also closes, and in closed form.

> **Theorem 2.** Let $N_{s,j}$ count the cells in state s with inheritance depth j. Then the entry of a line p **distributes** the old cells as follows — each line below reads "an old cell of the named state and depth j contributes to": $NN_j \to (p-2)NN_j + NO_{j+1} + ON_{j+1}$, $NO_j \to (p-2)NO_j + NO_{j+1} + OO_{j+1}$, ON likewise, $OO_j \to (p-2)OO_j + 2OO_{j+1}$. (3.1)
> Equivalently, marking a strike by x and the two rails by u and v, the full census is read off from $\boxed{\prod_{5\le q\le z} ((q-2) + xu + xv).}$ (3.2)

Proof. Of the p copies of any cell, exactly two are struck — one on each rail — by [P2, Thm 3]. A copy that is struck acquires one further dividing line, hence depth $j+1$; the other $p-2$ copies retain depth j. Reading off which rails are affected in each state gives (3.1), and (3.2) is the generating-function form of the same statement, one factor per line. ■

Verification. Zero error at $p = 7, 11, 13, 17, 19$, over full cycles up to $M =$ 1,616,615 cells. For the lines up to 19, expanding (3.2) gives

$$NN = 378675 = \prod (q-2),$$

$$NO = ON = 325980x + 106299x^2 + 17000x^3 + 1425x^4 + 60x^5 + x^6,$$

$$OO = 212598x^2 + 102000x^3 + 19950x^4 + 1800x^5 + 62x^6,$$

each matching the direct census exactly. Setting $x = 1$ recovers Theorem 1.

Two consequences.

1. **The state space is finite at every depth**, since $j \le \pi(z)$. One does not need to track a continuous weight; the full distribution of depth is a vector of length $O(\pi(z))$, and it transports exactly.
2. **Hence any weight of the form $w = f(\Omega_{\le z})$ is computable exactly on the cycle, for an arbitrary f — including a truncation such as $\max(0, \cdot)$.** Nonlinearity is harmless once one carries the distribution rather than its mean. The restriction to be kept in mind is that x marks *that* a line struck and not *which*: a weight depending on the sizes of the factors is not of this form. We return to both points in §4.

A third channel, and where the missing strike goes. Of the p residues of a line, (3.2) marks two — one per rail — and gathers the remaining $p-2$ as inert. But a line has a third strike inside every unit of the grid: the position $6j+3$, which lies on L_3 and is therefore already closed. Marking that strike by a variable w of its own splits the inert block and gives

$$\prod_{5\le q\le z} ((q-3)+u+v+w). \qquad (3.3)$$

This is (3.2) with x set to 1 and $q-2$ resolved into $(q-3)+w$; the cell counts are unchanged, so it refines Theorem 1 rather than correcting it.

> **Corollary 1.** The number of cells struck on neither rail is $\prod(q-2)$ as before, and its splitting by the number k of lines that spend their strike on the L_3 position is read off from $\prod((q-3)+w)$.

Verification. Expansion of (3.3) reproduces the full census — every triple (left strikes, right strikes, L_3 strikes) — for the line sets $\{5,7\}$, $\{5,7,11\}$ and $\{5,7,11,13\}$, with no discrepancy. For $\{5,7\}$ the open cells number $15=8+6+1$, which is the expansion of $(2+w)(4+w)$; growing the set gives $3, 15, 135, 1485, 22275$ open cells with the factor $q-2$ at every step, and the leading term $\prod(q-3)$ counts those where no line spent a strike on L_3 at all.

The channel carries no information about primality, and this is worth stating so it is not looked for. Among the cells left open by the lines 5 through 19, the proportion that are twin pairs, taken separately for each value of k, is 0.996, 1.010, 0.984 and 0.978 times the overall rate — every ratio within three per cent of one, over 234,237 cells. The refinement is a finer bookkeeping of the same residues, not a new discriminant.

The same test one layer up. [P2, Cor 2] reads the fate of a cell C_j through the phase $u_p(j)=jc^{-1}$, closure being $u=\pm1$. The centre of the diamond of two cells C_a, C_b is the cell of index $6ab$, whose two members are $36ab\mp1$; so the centre is closed to p exactly when $36ab\equiv\pm1$. Since $c^{-2}\equiv 36 \pmod p$,

$$u_p(a)\cdot u_p(b) = ab\cdot c^{-2} = 36ab,$$

so the gate applies directly to the **product of the two phases**, with no sign to track: the centre is closed to p if and only if $u_p(a)u_p(b)=\pm1$. The same two gates therefore decide one layer up, and the local count is again exact.

(A caution for anyone re-deriving this. The product $u_p(a)u_p(b)$ is $36ab$, which is not the phase of the centre cell: writing $p=6c+\sigma$ gives $c^{-1}\equiv-6\sigma$ and hence $u_p(6ab)=6ab\cdot c^{-1}=-\sigma\cdot 36ab$, the product only up to the sign $-\sigma$. Nothing depends on this, because the closure test above is on $36ab$ itself; but the two quantities are not the same, and at $p\equiv1 \pmod 6$ they differ.)

> **Corollary 2.** Of the $(p-2)^2$ ordered pairs of phases open to a line p, exactly $2(p-3)$ have product ±1. Hence the number of open pairs whose diamond centre is also open to p is $(p-2)^2-2(p-3)=(p-3)^2+1$.

Proof. By the identity above the centre is closed to p exactly when $u_p(a)u_p(b)=\pm1$. For $u\neq0,\pm1$ the two dangerous partners are u^{-1} and $-u^{-1}$; both are open, since $u^{-1}=\pm1$ would force $u=\pm1$, and they are distinct because p is odd. There are $p-3$ such u, and $u=0$ has no dangerous partner. ■

Verified: $5, 17, 65, 101, 197, 257$ at $p = 5, 7, 11, 13, 17, 19$, by direct enumeration; and the identity $u_p(a)u_p(b) = 36ab$, together with the equivalence between closure of the centre and $u_p(a)u_p(b) = \pm 1$, was checked over every ordered pair of residues for every prime below 60.

A caution that belongs with it. The count is **conditional** — it is a count over pairs of cells already open to p, not a density on the cycle — and the phase computation nowhere uses primality, so like the third channel above it is finer bookkeeping of residues. In particular a law of the shape "the centre dies unless $p \mid ab$" is false as stated: the birth index of [P2, Cor 1] is an exception, and at $p = 5$ it accounts for every one of the 49 failures among 11,175 pairs of twin indices.

A negative control worth recording. The obvious cheaper refinement — binning cells by the *value* of a log-weight into a fixed number of bins — does **not** close: prediction error 0%, 24%, 14%, 10% at $p = 7, 11, 13, 17$. The exactness depends on refining by a quantity that changes by exactly one under a strike, which the depth j does and a binned weight does not.

---

4. Weights on the cycle

4.1 What Theorem 2 gives, and what it does not

Weighted sieves attach to each surviving element a weight depending on its factorisation, and deduce an almost-prime from the positivity of the weighted sum. The weight that powers the current explicit results between consecutive squares and cubes (see [P10, §2.5] for those) is Richert's [2]:

$$w(a) = \lambda - \sum_{\substack{p^\alpha \| a \\ z \le p < y}} \alpha\Big(1 - \frac{\log p}{\log y}\Big), \qquad \lambda = k + 1 - k_2,$$

with $z = X^{1/k_1}$ and $y = X^{1/k_2}$; the mechanism is that $w(a) > 0$ forces $\Omega(a) \le k$, so a positive lower bound for $\sum w(a)$ over the sifted set produces an almost-prime. (The constant is λ, not 1; with $\lambda = 1$ the weight detects $\Omega \le k_2$ instead.)

The multiplicity α is part of the weight and not a decoration, and we write it out because dropping it breaks the detection: summed over *distinct* primes instead, the number $a = 5^6$ at $z = 4$, $y = 100$, $k = 3$, $\lambda = 1$ carries the single term $1 - \log 5/\log 100$, so $w(a) = \log 5/\log 100 > 0$ while $\Omega(a) = 6$. The mechanism counts prime factors with multiplicity because Ω does. The cycle modulo $\prod q$ cannot see a square of a line, so Theorem 2 counts distinct lines. The multiplicity is recovered on the cycle modulo $\prod q^e$: there each line contributes the factor

$$(q^e - 2q^{e-1}) + \sum_{a=1}^{e-1} (q^{e-a} - q^{e-a-1})x^a(u + v) + x^e(u + v),$$

with x^a marking a member divisible by exactly q^a (and x^e by q^e or more). The proof is that of Theorem 2, by the Chinese remainder theorem; $e = 1$ gives back $(q - 2) + xu + xv$, and once q^e exceeds the window the exponent is the true multiplicity, which is what Richert's α needs. Checked against a direct census on the full cycles modulo $25 \cdot 49$ and $25 \cdot 49 \cdot 121$.

It is a nonlinear function of the factorisation and, through $\log p/\log y$, of the *sizes* of the factors.

It is tempting to conclude that the framework cannot carry such a weight, since its objects are binary or counts. **That conclusion is half false, and Theorem 2 says which half.** The refined

law transports the *full distribution* of inheritance depth, in a finite state space of size $O(\pi(z))$, exactly. Consequently:

> **Corollary 3.** On the full cycle, $\sum_{\text{cells}} f(\Omega_{\le z})$ is computable exactly for an arbitrary function f, including a truncation $\max(0,\cdot)$, by reading the coefficients of (3.2).

So nonlinearity is harmless once one carries the distribution rather than its mean. The earlier obstruction we recorded — that a truncation cannot be tracked — was an artefact of following only the four totals.

But Ω alone is not enough for Richert. The weight above is not a function of $\Omega_{\le z}$: through $\log p/\log y$ it depends on the *sizes* of the prime factors, whereas the marker x in (3.2) records only *that* a line struck, never which. Four numbers with the same count make the gap concrete: $143,\ 2167,\ 2483,\ 323$ each have exactly two prime factors in $(10, 200)$, and their weights at $y = 200$ are $-0.0633,\ 0.4497,\ 0.4754,\ 0.0905$.

4.2 Why binning by the *value* of the weight does not work

The cheap repair is to bin cells by the value of a log-weight into a fixed number of bins and carry the bin counts. It does **not** close: prediction error 0%, 24%, 14%, 10% at $p = 7, 11, 13, 17$. The reason is that a cell's actual weight inside a bin is unknown at the moment $\log p$ is added, so the update is not determined by the state. Exactness needs a refinement by a quantity that changes by **exactly one** under a strike — which the depth does and a binned weight does not. The correct repair therefore refines the *line*, not the *weight*.

4.3 Refining the generating function by the size of the line

The obstruction identified in §4.1 is that the marker x in (3.2) records *that* a line struck and never *which*. The repair is to mark by size.

> **Corollary 4.** Let β assign to each line a bin. Then on the full cycle $\displaystyle\prod_{5\le q\le z}\big((q-2)+u_{\beta(q)}+v_{\beta(q)}\big)$ (4.1) transports the **joint distribution of the per-bin strike counts on each rail**, exactly.

Proof. Identical to Theorem 2. Of the p copies of any cell exactly two are struck, one on each rail; the only change is that the markers attached to the striking line are $u_{\beta(q)}$ and $v_{\beta(q)}$ rather than xu and xv. ■

Two variables per bin are needed, and one is not enough. Writing the factor as $(q-2)+x_{\beta(q)}u+x_{\beta(q)}v$ — one bin marker carrying both rails — loses exactly the joint distribution the corollary claims: a left strike from bin 1 with a right strike from bin 2 and the reverse configuration both produce the monomial x_1x_2uv, and no coefficient separates them. That form was printed in earlier versions of this section. It transports the per-bin totals and the per-rail totals; it does not transport their joint law, which is what a weight evaluated on one member of a cell needs.

Verification. Full cycle for the lines $5, 7, 11, 13, 17, 19$ (M = 1,616,615 cells), binned by size into equal blocks: the census predicted by (4.1) agrees with the direct census in **every** state and with no error — 100 states at $B = 2$ and 216 at $B = 3$, which are exactly the $\prod_b \binom{n_b+2}{2}$ of the count below. The one-marker form has only 64 and 108 distinct monomials for the same line set, which is the collapse just described. Regenerated by `code/verify_bin_generating_function.py`.

The cost is polynomial in $\pi(z)$ for a fixed number of bins, and only then. The state space has size $\prod_b \binom{n_b+2}{2}$, where n_b is the number of lines in bin b — a polynomial of degree $2B$ in $\pi(z)$, so the degree grows with the resolution and the table below is a statement about fixed B, not a claim that the refinement is cheap at every resolution. The bins in the table are equal in $\log p$:

z	lines	depth only (Thm 2)	$B = 2$	$B = 4$	$B = 8$	unrefined joint, $2^{\pi(z)}$
1,000	166	167	1.9×10^6	9.1×10^9	3.3×10^{15}	$\approx 10^{50}$
20,000	2,260	2,261	5.0×10^9	8.2×10^{15}	6.5×10^{26}	$\approx 10^{680}$

And the resolution Richert's weight requires is modest. Measured on 4×10^6 cells with $X = 2.4 \times 10^7$ and the parameters $k = 3$, $k_1 = 8$, $k_2 = 3.17$ (so $z = 8.4$, $y = 212.9$, $\lambda = 0.83$, and 43 lines in the weight range), comparing $\sum \max(0, w)$ computed from the binned counts against the exact value:

bins B	1 (depth only)	2	4	8	16
relative error	+40.5%	+8.2%	+2.0%	+0.39%	+0.10%

The +40.5% in the first column is exactly the error of ignoring the sizes, i.e. the gap left by Corollary 3; eight bins close it to 0.4% **on that experiment**. It is a measured accuracy at one height and one parameter set, not a uniform error bound: what the bins control is the resolution of $\log p / \log y$ inside a bin, so the error is governed by the bin width and by how many factors fall in the wide bins, and neither is estimated here.

The *factor-size* deficit is removable: exact at every resolution, polynomial in $\pi(z)$ for fixed B, 0.4 per cent at B

5. Ownership layers

This section changes the bookkeeping object. Sections 2–4 count **cells** and therefore carry the two-rail factor $1 - 2/q$. Here we assign an owner to each **individual odd integer**, so the corresponding one-point density is $1 - 1/q$. The ownership decomposition is exact on a full CRT cycle and removes inclusion–exclusion by construction.

5.1 Ownership, and disjoint layers

Assign to each odd n its **owner**, the smallest line striking it:

$$O(n) = \min\{q : q \mid n, q \neq n\}, \qquad O(n) = N \text{ if none.}$$

Thus $O(25) = O(35) = 5$, $O(45) = 3$, $O(49) = 7$, and $O(105) = 3$ — priority resolves every overlap, so 105 belongs to 3 alone although 5 and 7 also divide it.

Theorem 3. Let $S_{<q}$ denote the survivors of all lines below q. Then the ownership layers $E_q = q \cdot S_{<q}$, taken from q^2 onward, are pairwise disjoint, and consequently $\mathrm{card} S(I) = \mathrm{card} I - \sum_q \mathrm{card} E_q(I)$ **with no inclusion–exclusion corrections.**

Proof. If n has owner q then $n = qr$ with r divisible by no line below q, i.e. $r \in S_{<q}$; and $n \geq q^2$ since $r \geq q$. Conversely any qr with $r \in S_{<q}$ and $qr \geq q^2$ has least prime factor q. So E_q is exactly $\{n : O(n) = q\}$, and distinct owners give disjoint sets. ■

Verification. $E_5 = \{25, 35, 55, 65, 85, 95, 115, 125, \dots\}$ and $E_7 = \{49, 77, 91, 119, 133, 161, 203, 217, \dots\}$ agree exactly, term by term, with $\{n : O(n) = 5\}$ and $\{n : O(n) = 7\}$.

A genuine structural gain, and a precise accounting of what it costs. The $2^{\pi(z)}$ intersection terms vanish by construction — the difficulty does not disappear, it moves: evaluating $\mathrm{card} E_q(I)$ requires knowing how many earlier survivors the layer copies, i.e. the *shape* of $S_{<q}$. **Each layer is a scaled copy of the previous survivor strip**, which is the same statement as the observation that all lines read a common word on a common strip.

5.2 Why disjointness repairs the union bound but not the estimate

The layering of §5.1 was reached by asking a specific question: **the union bound over lines fails because it double-counts, so does the failure disappear once the layers are genuinely disjoint?** The answer is instructive, and worth recording because the first half of it is a real gain.

The union bound. Each line q strikes at most a fraction $1/q$ of an interval, so the naive bound on the struck fraction is $\sum_q 1/q$. Over odd $q \leq z$ this diverges like $\log\log z$ and exceeds 1 almost immediately — on cells, where a line removes two classes and the sum is $\sum 2/q$ over $q \geq 5$, a window of 100 cells is already spent by the fourth line ($40 + 29 + 18 + 15 = 102$); on the odd integers it passes 1 at $q = 29$. **This failure is caused entirely by double-counting: a number divisible by both 3 and 5 is charged twice.**

The layered count. By Theorem 3 the layers are disjoint, so no position is charged twice. On a full CRT cycle the struck fraction is exactly $\sum_q |E_q|/|I|$; on a shorter interval the same disjoint identity holds for counts, but the simple product formula below acquires boundary dependence. Comparing the full-cycle densities:

z	union bound $\sum_q 1/q$	layered bound $\sum_q \lvert E_q\rvert/\lvert I\rvert$	survivors
13	0.844	0.616	0.384
101	**1.313**	0.762	0.238
997	**1.698**	0.838	0.162
10^6	**2.387**	**0.919**	0.081

The layered bound is below 1 for every z, however many lines are summed.

So the summation failure is genuinely repaired: disjointness is exactly the property the union bound lacked.

But on the full cycle the repaired sum returns the sieve product and nothing more. Since each residue class occurs exactly once in every q copies, $|E_q| = \frac{1}{q}|S_{<q}|$ on the full CRT cycle; the layer sizes already contain the product of everything below, and

$$\sum_{q \leq z} \frac{|E_q|}{|I|} = 1 - \prod_{q \leq z} \Big(1 - \frac{1}{q}\Big), \qquad \text{hence} \qquad \frac{|S|}{|I|} = \prod_{q \leq z} \Big(1 - \frac{1}{q}\Big).$$

Verification. Measured layer fractions in $(P^2, 9P^2)$ at $P = 997$: $|E_3| = 0.3333$, $|E_5| = 0.1333$, $|E_7| = 0.0762$ — matching 1/3, 2/15, 8/105 to the four decimals shown, and the same at $P = 101$ and $P = 499$.

The interference did not disappear; it moved. It is no longer in the intersections — it is inside the size of each layer. Three routes therefore reach the same place:

route	result
naive union bound	diverges
disjoint layers	$\prod(1 - 1/q)$
inclusion–exclusion	$\prod(1 - 1/q)$

The layered form is the cleanest of the three and the only one whose bound never exceeds 1, but it is not stronger. Evaluating $|E_q(I)|$ requires $|S_{<q}|$ on the interval I/q — the same problem one level down. That recursion is Legendre's, and its shape and its limitations are classical.

---

6. The correlation ladder: transport of the second moment

Sections 2–4 transport counts. This section transports *pairs of counts*, and the outcome settles a question the earlier sections leave open: what is the smallest object that closes under the entry of a new line, if one wants variances and not only means.

6.1 The pair indicator and its autocorrelation

Index gap-2 pairs by x, the pair being $(2x+1, 2x+3)$, and set $A(x) = 1$ when both members survive the line set, 0 otherwise. A is periodic with period $M = \prod q$, and $\sum_x A(x) = T = \prod(q-2)$. Define the autocorrelation

$$C(h) = \sum_{x \bmod M} A(x)A(x+h).$$

Theorem 4. Under the entry of a line q, $C_{\text{new}}(h) = K_q(h)C_{\text{old}}(h)$, $K_q(h) = \begin{cases} q-2, & h \equiv 0 \pmod q, \\ q-3, & h \equiv \pm 1 \pmod q, \\ q-4, & \text{otherwise,} \end{cases}$ and hence $C_Q(h) = \prod_{q \in Q} K_q(h)$ outright, with no cycle of length M ever being built.

Proof. For a single line q, the pair at x dies exactly when $2x + 1 \equiv 0$ or $2x + 3 \equiv 0$, i.e. when $x \in \{c, c-1\}$ with $c \equiv -1/2 \pmod q$ — **two consecutive classes**, and that is what makes the ladder descend in steps of one. For the pairs at x and at $x + h$ both to survive, x must avoid

$$\{c,\ c-1\} \cup \{c-h,\ c-1-h\},$$

a union of size 2, 3 or 4 according as $h \equiv 0$, $h \equiv \pm 1$, or otherwise modulo q. Multiplicativity across the lines is the Chinese remainder theorem. ■

Verification. Compared with the definition over the full cycle for the line sets $\{3,5\}$, $\{3,5,7\}$, $\{3,5,7,11\}$ and $\{3,5,7,11,13\}$ — periods up to $M = 15{,}015$ — at every h from 0 to 39: exact in every case. For the last set, $C(0),\dots,C(8) = 1485, 0, 0, 189, 0, 0, 504, 0, 0$.

A corollary visible in that data. $K_3(h) = 0$ whenever $h \not\equiv 0 \pmod 3$, because L_3 leaves a single class modulo 3. Hence

$$C(h) = 0 \quad \text{unless} \quad 3 \mid h,$$

and the surviving pairs live on a sublattice. This is the effect of the grid line isolated in its cleanest form.

The five rungs of the ladder now all have meanings: q counts cells, $q-1$ single survivors (§2), $q-2$ one pair, $q-3$ two pairs at distance ± 1, and $q-4$ two pairs in general position.

6.2 Why no single number closes the second moment

For a window of L consecutive pair slots placed at s, write $N_s(L) = \sum_{j<L} A(s+j)$. Averaging over s gives the mean LT/M and, exactly,

$$\operatorname{Var}(N_L) = \frac{LT + 2\sum_{h=1}^{L-1}(L-h)C(h)}{M} - \Big(\frac{LT}{M}\Big)^2,$$

with no sampling and no independence assumption anywhere.

> **Corollary 5.** For windows of length $L \geq 4$ the variance after the entry of q is not a fixed multiple of the variance before it: it depends on how $C(h)$ is spread over the residue classes of h modulo q, which the variance does not record.

Proof. By Theorem 4 the entry of q multiplies $C(h)$ by a factor depending on h mod q, so Var' is a different linear functional of C from Var. That the ratio is not fixed is shown by one exact instance from the formula above: at $L = 6$, the entry of 13 multiplies the variance by $\frac{1104}{4225}/\frac{4}{25} = 1.633$ for the line set $\{3,5\}$, and by $\frac{13164}{41405}/\frac{72}{245} = 1.082$ for $\{3,5,7\}$. ■

The hypothesis $L \geq 4$ is needed: with the line 3 present, $C(1) = C(2) = 0$, so for $L \leq 3$ the variance is $x - x^2$ with $x = LT/M$, and the entry of q only rescales x by $(q-2)/q$.

A related measurement, not the same quantity. The variance of the deviations across square-anchored windows, for the line sets $\{3\}$, $\{3,5\}$, $\{3,5,7\}$, $\{3,5,7,11\}$, is 0.0741, 0.1689, 1.1316, 2.2880 — successive ratios 2.28, 6.70, 2.02, with no pattern — in line with Corollary 5, though these are windows anchored at squares, not the cycle average of the formula.

So the minimal object that closes the second moment is the function $C(h)$, not a number. The four-state census of §2 is exact because a single window count is a linear functional of A; a variance is a quadratic one, and quadratic functionals need the correlation itself.

6.3 The general pattern, and what it is

Nothing above used h being a single displacement. For any finite pattern of displacements H, the same argument gives

$$N_{\text{new}}(H) = (q - \nu_q(H))N_{\text{old}}(H),$$

where $\nu_q(H)$ is the number of distinct residues modulo q that the pattern forbids. Theorem 4 is the case $|H| = 2$ and §2 the case $|H| = 1$.

This should be named for what it is. The resulting product $\prod_q(1-\nu_q(H)/q)$, suitably normalised, is the singular series of the Hardy–Littlewood k-tuple conjecture for the pattern H [1, Ch. 1]. The transport law is therefore not a new object; what §6.1 adds is that the framework produces it by a direct combinatorial count on the cycle, with the ladder $q-2, q-3, q-4$ read off from the geometry of two consecutive forbidden classes rather than assembled from local densities.

And the limitation is the same as everywhere in this paper. These laws describe the distribution over *all* phases of the cycle. They say nothing about any one designated phase, and a designated phase is what an application needs.

---

7. What is exact here, and what is not

Exact on the cycle, with proof: the four-state update and its closed solution (Theorem 1); the refined law and its generating function (Theorem 2); arbitrary weights in $\Omega_{\le z}$ (Corollary 3); the refinement by line size at any resolution (Corollary 4); the disjointness of the ownership layers and the resulting inclusion–exclusion-free count (Theorem 3); and the transport of the autocorrelation with the ladder $q-2, q-3, q-4$ (Theorem 4), together with the consequence that no single number closes the second moment (Corollary 5); and, on the cycle modulo $\prod q^e$, the census by multiplicity (§4.1).

Arithmetic, not measurement: the cost table of §4.3, which counts states.

Measured, not proved: the iteration of (2.1) to $P = 2\times 10^8$ (§2.3); the bin-resolution table of §4.3 and the 24% negative control of §4.2; the layer fractions of §5.2; and the variances across square-anchored windows in §6.2.

Not addressed here at all: every theorem above is exact over a full cycle of length $\prod q$. Applications need a window of length $\asymp z^2$. Paper 4 measures that transfer separately. On the tested windows, soft depth weights transfer very accurately, while the sharp depth-zero indicator shows the familiar deficit near 0.80. Those are window measurements, not consequences of the exact cycle identities proved here.

No new bound and no improvement to any known result is claimed. The survivor density $\prod(1-2/q)$ is the classical $V(z)$ of a sieve of dimension $\kappa = 2$; Theorem 1 dissolves into Mertens products and contains no new constant; Theorem 3 is the least-prime-factor decomposition, and the recursion of §5.2 is Legendre's. What is offered is that several quantities usually carried as estimates are, in these coordinates, identities — and, in the case of weights, that the identity survives a refinement fine enough to be useful.

No progress toward the twin-prime conjecture is claimed, and no new bound. Priority is not claimed for any result.

The computations and much of the prose in this paper were prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running the computations, and for auditing the papers against their own scripts. All statements were checked by the author, who is responsible for them; the repository README sets out the division of labour in full.

References

The companion papers of this set are cited as [P1] to [P12], and the numbered entries below are the external works. The two kinds never share a number: a bracket with a P is a companion paper, a bare number is a reference in the list below. This paper imports only the definitions and statements of the companion papers, never their proofs.

1. J. Friedlander and H. Iwaniec, *Opera de Cribro*, AMS Colloquium Publications **57**, 2010. — *the Mertens products of §2.3 and the standard form of the weighted sieve against which Corollary 4 is compared.*
2. H.-E. Richert, *Selberg's sieve with weights*, Mathematika **16** (1969), 1–22. — *the weight $w(a)$ discussed in §4; it depends on the sizes of the prime factors, which is precisely what Corollary 3 does not transport and Corollary 4 repairs.*